Massey University
College of Sciences

161.304: Computer Laboratory 11

Exercise 1: Rejection Sampling for a Posterior from the Insecticide Data

We met the insecticide data in Lab 8. For treatment B2, $x = 41$ insects were killed out of a total of $n = 50$. We will model x as a $\mathsf{Bin}(n, p)$ random variable.

Suppose that the prior density for p is

$$f(p) = \begin{cases} \frac{\pi}{2}\sin(\pi p) & \text{if } 0 \le p \le 1 \\ 0 & \text{otherwise.} \end{cases}$$

Plot this prior density. (You might like to note that the value of π is a predefined object in R; just type `pi` and hit return to see its value.)

By following a suitable Example from notes or lectures (or just by being naturally brilliant!) write code to generate $S = 10,000$ simulations from the posterior distribution for p. Create a histogram from these posterior simulations. Calculate the (approximate) posterior mean for p, and a 95% credible interval.

Exercise 2: Monte Carlo Integration

Let the random variable X have probability density function f. The expected value of $g(X)$ for some function g is given by

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} f(x)g(x)\,dx \tag{1}$$

Thanks to the Law of Large Numbers, we know that this can be well approximated by the sample mean

$$\frac{1}{S}\sum_{i=1}^{S} g(x_i)$$

where $x_1, \ldots, x_S$ is a random sample from f.

It follows that we can get a good approximation to any integral of the form of equation (1) using simulation. Consider, for example, the integral

$$I = \int_0^{\infty} x\sin(x)e^{-3x}\,dx\ .$$

This can be rewritten as

$$I = \int_0^{\infty} \left(3e^{-3x}\right)\left(\tfrac{x\sin(x)}{3}\right)\,dx$$

and is hence of the form of equation (1) with $g(x) = x\sin(x)/3$ and $f(x)$ being an $\mathsf{Exp}(3)$ density function.

Implement the following in R. For a simulation size $S = 100,000$, generate a random sample $x_1, \ldots, x_S$ from the $\mathsf{Exp}(3)$ distribution. Compute $g(x_i) = x_i\sin(x_i)/3$ for each simulation, and then compute the mean $\frac{1}{S}\sum_{i=1}^{S} g(x_i)$. This will be a (very close) approximation to the integral I, as you will be able to see if you calculate a suitable estimate of the precision of your estimate.

This simulation approach to evaluating integrals is called *Monte Carlo integration*.

Exercise 3: A Simple MCMC Sampler

Consider the distribution $f(\theta)$ with probability mass function

$$f(\theta) = \begin{cases} \frac{9-\theta}{36} & \text{if } \theta = 1, 2, 3, 4, 5, 6, 7, 8 \\ 0 & \text{otherwise.} \end{cases}$$

In this Exercise we will write a Markov chain Monte Carlo (MCMC) sampler to generate simulations from this distribution. (Of course, we could sample from it directly, so this is purely for illustrative purposes.)

The following algorithm provides pseudo code for a suitable MCMC sampler.

Set initial value $\theta^{(0)}$; set $t = 1$
for $t = 1 : S$ **do**
  Sample candidate $\theta^\dagger$ from $\{1, 2, 3, 4, 5, 6, 7, 8\}$ with equal probabilities.
  Compute acceptance probability $\alpha = \min\left\{1, \frac{f(\theta^\dagger)}{f(\theta^{(t)})}.\right\}$
  Sample U from $\mathsf{Unif}[0, 1]$
  if $U \leq \alpha$ **then**
    Set $\theta^{(t+1)} = \theta^\dagger$
  else
    Set $\theta^{(t+1)} = \theta^{(t)}$
  end if
end for

Implement this code in R, for $S = 11,000$ iterations (after the initial value). Produce a line plot of the last 1000 simulated values against iteration number. (Use the `plot` command with argument `type="l"`.)

Discard the first $1,001$ values (i.e. $\theta^{(0)}, \ldots, \theta^{(1000)}$) as a burn-in period, affected by the choice of initial conditions. Save the last $10,000$ iterations as an R vector `theta.sim`. Compute the relative frequencies of the possible outcomes $\{1, 2, 3, 4, 5, 6, 7, 8\}$ by

```
table(theta.sim)/10000
```

Compare these relative frequencies with the probabilities given by $f(\theta)$. Has your MCMC sampler produced simulations in the correct proportions? Try repeating the sampling process with $S = 1001000$ iterations.